VIBRATION—DRIVELINE DRONE OR VIBRATION—FELT AT 80-96 KM/H (50-60 MPH)—VEHICLES EQUIPPED WITH 228 INCH WHEELBASE AND 6.8L ENGINE ONLY	Article No. 01-17-3

FORD: 1999-2002 SUPER DUTY F-53 STRIPPED CHAS.

ISSUE

Some Super Duty F53 Motor Home Stripped Chassis vehicles built with a standard 228 inch wheel base may exhibit a driveline drone/vibration condition. This drone has a frequency of 80 Hz to 120 Hz and is most noticable at constant speed or on light acceleration between 80-96 km/h (50-60 mph). This may be caused by an excessive engine/transmission operating angle.

ACTION

Verify the engine/transmission operating angle is set to the specified angle of 5 degrees. If the engine/transmission operating angle is excessive add shims between the transmission mount and the transmission. Shim housing to raise the rear of the transmission until the 5 degrees is indicated at the engine oil pan rail. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

Vehicle Test Drive:

Test drive the vehicle to verify that the condition is a second order driveline vibration. Conduct this evaluation on a smooth road and use an EVA (Electronic Vibration Analyzer) to insure the correct source of the vibration condition. Refer to TSB 99-11-01 or section 100-04 of the appropriate year F-Super Duty/Excursion/F53 Motorhome Workshop Manual for a description on using the EVA. If the driveline vibration has been verified, proceed as follows:

1. Measure the engine/transmission operating angle (as taken from the engine oil pan rail) using an Angle Master. The angle should be approximately 5.0 degrees.
2. If the engine/transmission operating angle exceeds 5.0 degrees, remove the fasteners from rear of transmission to transmission mount.
3. Install spacers (1C3Z-7C144-AA) between the transmission and the transmission mount to acquire a new driveline angle of 5.0 degrees. Each spacer is worth approximately 0.5 degrees.
4. If two or three spacers are required, use the new fastener 390914 S426 (2 inch bolt) and torque to 90 +/- 13 N•m (66 +/- 9 lb-ft). BE SURE to check the depth of the hole in the transmission case before installing the new fastener.

NOTE

DO NOT INSTALL AN EXCESSIVELY LONG BOLT, AS THIS WILL CRACK THE TRANSMISSION CASE.

5. Test drive to verify vibration/drone is corrected.

PART NUMBER	PART NAME
1C3Z-7C144-AA	Driveline Spacer
390914 S426	Bolt

OTHER APPLICABLE ARTICLES: 99-11-1
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

OPERATION	DESCRIPTION	TIME
011703A	Measure Engine And Transmission Angle And Adjust As Necessary (Includes Time For Test Drive Before And After Repair)	1.1 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
5025	D9

OASIS CODES: 597997, 702000, 703000, 703200, 703300

Copyright © 2001 Ford Motor Company